

E04 CSTR Energy Balance Example

The elementary liquid phase reaction $A \rightarrow B$ is carried out in a non-isothermal CSTR. Species A and an inert I are fed to the reactor in equimolar amounts. The molar feed rate of A is 80 mol/min.

The heat capacity of the inert is 30 cal/mol/°C. The heat capacities of A and B are 20 cal/mol/°C.

$$\tau = 100 \text{ min}$$

- assume irreversible

$$\Delta H^{rxn} = -7500 \text{ cal/mol}$$

$$UA = 8000 \text{ cal/min/°C.}$$

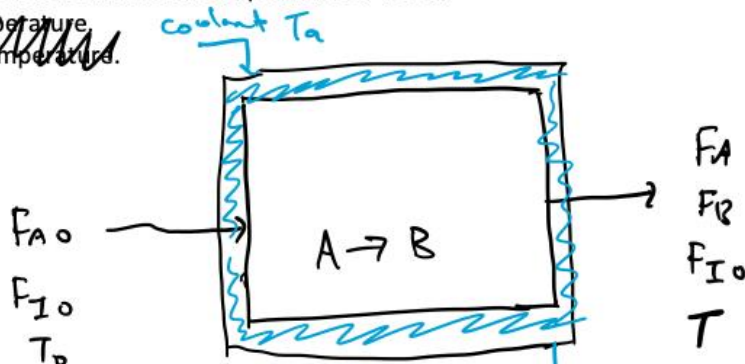
$$T_a = 300 \text{ K}$$

$$k = 0.0066 \text{ min}^{-1} \text{ at } 350 \text{ K}$$

$$E = 40 \text{ kcal/mol.}$$

Determine:

- The reactor temperature for a feed temperature of 450 K.
- The ignition temperature.
- The extinction temperature.



- develop the design equations:

① mole balance

② energy balance

coolant T_a
assume constant T_a

$$\underline{EB} \quad 0 = F_{A0} H_{A0} + F_{I0} H_{I0} - F_A H_A - F_B H_B - F_{I0} H_I + UA(T_a - T)$$

assume $\dot{W}_S = 0$

next, write F_A & F_B in terms of X

$F_A = F_{A0}(1-X)$, $F_B = F_{A0}X$. Sub these into the EB

$$0 = F_{A0} H_{A0} + F_{I0} H_{I0} - F_{A0}(1-X) H_A - F_{A0}X H_B - F_{I0} H_I + UA(T_a - T)$$

group like terms: sensible heat terms & heat of rxn terms
to do that I will distribute:

$$0 = F_{A0} H_{A0} + F_{I0} H_{I0} - F_{A0} H_A + F_{A0}X H_A - F_{A0}X H_B - F_{I0} H_I + UA(T_a - T)$$

$$0 = F_{A0}(H_{A0} - H_A) + F_{I0}(H_{I0} - H_I) + F_{A0}X(H_A - H_B) + UA(T_a - T)$$

sub in definitions:

$$H_i = \Delta H_{f,i}^\circ + \int_{298.15K}^T C_{pi} dT$$

$$\text{example: } \underset{\uparrow}{H_{A0}} - \underset{\uparrow}{H_A} = \Delta H_{f,A}^\circ + \int_{298.15K}^{T_0} C_{pA} dT - \Delta H_{f,A}^\circ - \int_{298.15K}^T C_{pA} dT$$

$$H_A(T_0) \quad H_A(T)$$

$$H_{A0} - H_A = \int_T^{T_0} C_{pA} dT. \text{ Here: } H_{A0} - H_A = C_{pA} (T_0 - T)$$

Since I've assumed my heat capacities are constant

$$\text{example: } H_A - H_B = \Delta H_{f,A}^\circ + \int_{298.15K}^T C_{pA} dT - \Delta H_{f,B}^\circ - \int_{298.15K}^T C_{pB} dT$$

$$H_A - H_B = -\Delta H_{rxn}^\circ + C_{pA} (T - 298.15K) - C_{pB} (T - 298.15K)$$

- totally fine to include $C_{pi} (T - 298.15K)$ terms ...

Also totally fine to assume ΔH_{rxn}° is constant w/ temperature

$$0 = F_{A0} C_{pA} (T_0 - T) + F_{I0} C_{pI} (T_0 - T) + F_{A0} X (-\Delta H_{rxn}^\circ) + UA (T_a - T)$$

rearrange for X :

$$X_{EB} = \frac{F_{A0} C_{pA} (T_0 - T) + F_{I0} C_{pI} (T_0 - T) + UA (T_a - T)}{(\Delta H_{rxn}^\circ) (F_{A0})}$$

$$\underline{MB} \quad V = \frac{F_{A0} X}{-r_A}$$

$$-r_A = k C_A$$

$$C_A = C_{A0} (1 - X)$$

(liquid phase rxn)

$$V = \frac{F_{A0} X}{k C_{A0} (1 - X)} = \frac{v_0 X}{k (1 - X)}$$

$$\gamma = \frac{x}{k(1-x)} \quad \cdot \text{ Solve for } x \circ$$

$$x_{MB} = \frac{\gamma k}{1 + \gamma k}$$

Simultaneously solve $x \in \mathbb{R} : x_{MB}$ for $x \in T$
